Student Self-Perception Bias Detection using Behavioral Logs and Facial Expression Analysis

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Introduction

This component detects when a student's self-perceived account of a study session — a diary entry — diverges from what is independently observable, and reports that divergence as both a labeled bias type and a 0–100 Perception Alignment Score (PAS).

Three signal families feed the detector. The **claimed** signal is the diary entry itself (activity category, claimed duration, claimed location, self-reported mood before/after). The **behavioral logs** signal is collected passively from the phone: categorized Android app-usage minutes (educational vs. entertainment vs. social media, with YouTube further split by on-screen video title), GPS matched against the student's saved reference locations, app-switch count, and an opt-in calendar overlap check. The **facial** signal is a short camera scan, read first by MediaPipe landmarks and then by a Vision Transformer emotion model, yielding a dominant emotion, a confidence score, and a stress indicator. A multi-signal comparator turns these three into 11 engineered features, which an XGBoost classifier (with a SHAP-style per-prediction explanation) and a separately-weighted PAS formula both consume; a small rule layer adds cross-check indicators the model alone might miss.

This directly answers both items of the final-presentation panel's PP1 feedback: that mood should be "set automatically from the other member's component" rather than self-selected (Section on the facial-to-diary hand-off, retained from the component's original scope), and that the component's "research value is limited" and needs "external evaluation" (addressed in the Evaluation Plan, which is explicit about what this system has and has not yet validated).

The component has three responsibilities:

1. **Signal collection** — behavioral logs (app usage, location, calendar) and a facial scan (emotion + stress), alongside the student's own diary entry.
2. **Comparison and classification** — an 11-feature comparator, an XGBoost multi-class bias classifier with SHAP-style explanations, a rule-based cross-check layer, and a weighted PAS score.

3. **Feedback to the student** — a bias type, a PAS score and level, a ranked explanation of which features drove the prediction, mood-fix recommendations when stress is elevated, and (via `pickerMode`) automatic mood hand-off into the diary-entry screen owned by a teammate's component.

Literature Review

Eight bodies of work ground this component: facial emotion recognition, Vision Transformer-based FER, facial stress detection, landmark-based affect analysis, mood-aware recommendation, cognitive self-report bias, behavioral-log vs. self-report accuracy, and explainable tabular ML.

Facial emotion recognition. Li and Deng's survey [1] frames deep FER as the dominant paradigm over handcrafted features. Wang et al. [2] catalogue eight open FER challenges — expression disturbance and compound emotions among them — that motivate this component's confidence-gated "Inconclusive" fallback. Liu et al. [3] show a lightweight self-difference CNN can isolate expression-relevant signal from identity, informing why low-confidence frames are treated as unreliable rather than forced into a class.

Vision Transformers for FER. The backend's emotion classifier follows the ViT line: Ezzameli and Mahersia [4] report classification rates above 97% on benchmark FER data, Li et al. [5] generate a neutral "poker face" to separate identity from emotion before classification, and Akeh and Chandra [6] fine-tune a ViT on a culturally local dataset, underscoring that FER accuracy is dataset-sensitive — carried into this project's evaluation plan.

Facial stress detection. Zhang et al. [7] detect stress from video rather than a single frame; Zhang et al. [8] fuse facial expression with voice and ECG. Both suggest this component's single-image `stress_indicator` is a simplification with known limits. A 2026 facial-emotion-aware CNN approach [9] reports 86% stress-assessment consistency against self-reported stress, used here as a rough benchmark. Explainable stress recognition via facial action units [10] motivates surfacing the dominant emotion and confidence alongside the stress score rather than the score alone.

Landmark-based affect analysis. MediaPipe's face mesh runs as a fast pre-check (face present, frontal, unblurred) before the ViT classifier. Mertes et al.'s AffectToolbox [11] and Mukhiddinov et al. [12] show landmark geometry carries usable signal even under partial occlusion, supporting this component's "partial face / blurry image" diagnostics rather than a silent failure. A broader MediaPipe Face Mesh evaluation [13] frames landmarks as a pre-processing and robustness layer, not a classifier replacement. Part-aware recognition work [14] further supports weighting face regions differently near the confidence threshold.

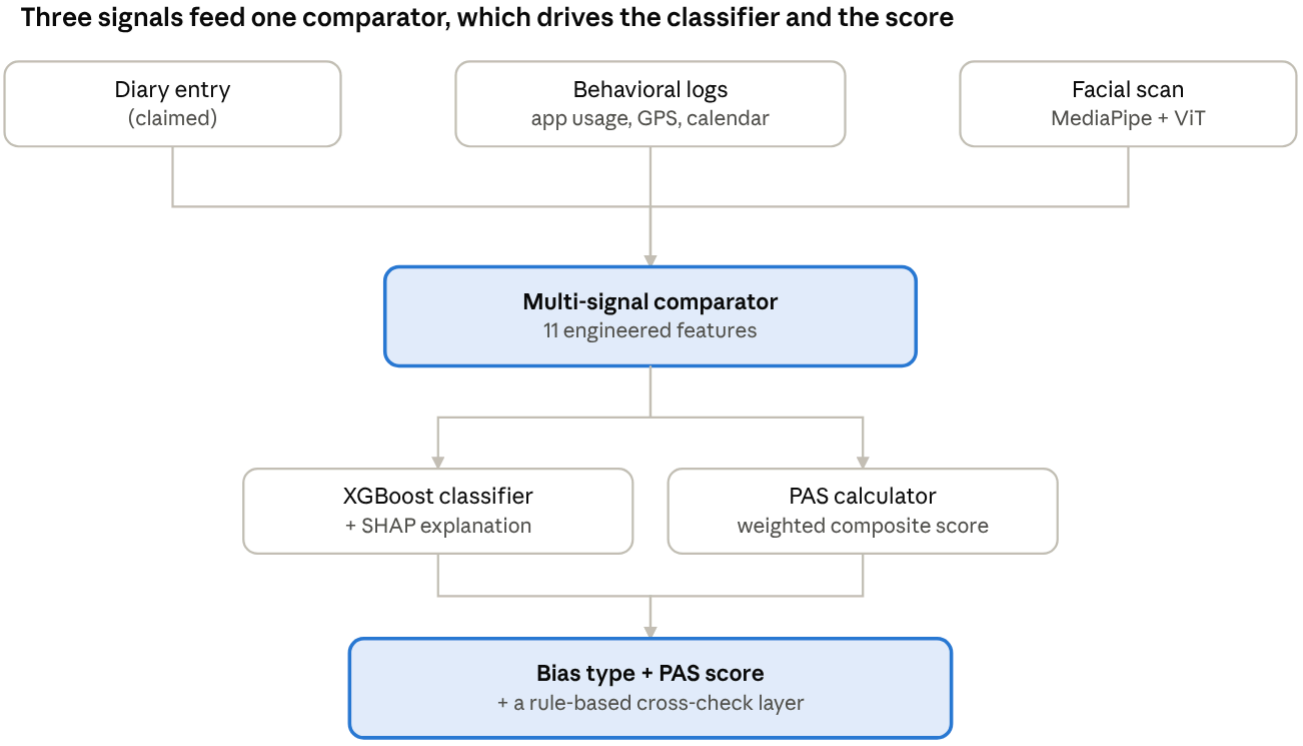
Mood-aware recommendation. Hasan and Bunescu's survey [15] distinguishes attitude-, emotion-, and mood-aware recommendation, placing this component's stress-triggered "tip" card in the mood-aware category — a slower affective state driving generic suggestions rather than a single dictated answer. Leung et al. [16] and mood-conditioned ranking work [17] describe the same pattern in text and music recommendation, which is why this component offers 2–3 cards rather than one fixed instruction.

Cognitive self-report bias. The panel's objection that a self-selected mood "won't be accurate" has a direct basis in the planning-fallacy literature. Kahneman and Tversky [18] showed people mis-estimate their own outcomes by anchoring on a best-case plan rather than past base rates; Buehler, Griffin, and Ross [19] found this persists even when people recall similar past tasks that ran long. Flyvbjerg [20] extends this to project-level self-assessment, arguing self-interested framing — not just forgetting — drives the estimate/outcome gap.

Behavioral logs vs. self-report. This is the empirical foundation for comparing a diary claim against a phone log at all. Boase and Ling [21] found self-report of phone use only moderately correlates with server log data; Ohme et al. [22] found adults under-report screen time by roughly 125 minutes a day on average against iOS Screen Time logs. Elevelt et al. [23] found GPS-derived locations substantially more reliable than diary-reported ones for deriving where someone actually was. Closer to this project's domain, Vaghela, Patel, and Shah [24] used app-usage and stress logging to distinguish "mindful" from "distracted" student phone behavior, and Zhao et al.'s ISense [25] predicts self-regulated learning from mobile-sensing logs across 211 students — both precedents for this component's behavioral feature set (app categories, switches, location, calendar).

Explainable tabular ML. The classifier is XGBoost [26], chosen for its strong accuracy on small, structured feature sets like this component's 11-column table. Each prediction is explained with Lundberg and Lee's SHAP framework [27] (via XGBoost's built-in `pred_contribs`), so a flagged entry shows which features drove it — required for a bias detector a student is expected to trust and act on, not just a bare label.

System Design



diary + behavioral logs + facial scan → comparator → classifier & PAS score

The facial-scan component from the original scope (automatic mood hand-off into the diary form) is one of three inputs here — it also feeds `facial_stress` into the comparator below and triggers the mood-fix recommendation tip when stress is high. The comparator's 11 features drive two parallel consumers: an XGBoost classifier (explained per-prediction with SHAP-style contribution weights) and a separately-weighted PAS score; a small rule layer then adds cross-check indicators the model alone might miss (e.g. a facial-stress signal the self-report didn't mention).

Implementation

Behavioral log collection (`SensorDataService` , Flutter/Android). App usage over the last few hours is pulled via `UsageStats` / `app_usage` and sorted into educational, entertainment, social-media and "other" minutes by a package-name lookup table; YouTube is a special case — an accessibility service reads the on-screen video title and a keyword list ("lecture", "tutorial", … vs. "vlog", "prank", …) reclassifies it rather than trusting the package name alone. GPS location is matched against the student's own saved reference points (Home, Library, …); calendar verification is opt-in and checks only for an overlapping event in the claimed time window, never raw calendar content.

Facial signal collection (`FacialCaptureScreen`). A front-camera photo is EXIF-normalized, then posted to `/api/bias/facial/analyze`, which runs a MediaPipe landmark check before a Vision Transformer emotion classifier, returning `dominant_emotion`, `confidence`, and `stress_indicator`. Below 0.35 confidence the component reports `Inconclusive` rather than a guess.

Multi-signal comparison (`MultiSignalComparator`). Claimed diary fields and verified sensor/facial data become 11 features (`duration_gap`, `duration_match_ratio`, `location_match`, `focus_quality_score`, `activity_match`, `stress_mismatch`, `app_switch_count`, `social_media_minutes`, `distraction_duration`, `calendar_match`, `facial_stress`). Duration and activity-match are only scored against verified educational app time for study-type categories (Lecture, Self-study, Assignment, Group work); for a leisure entry (e.g. "Play", "Gym"), 0 minutes of educational app usage is expected and truthful, so that signal is neutralized rather than flagged as a mismatch — a deliberate guard against false positives on honest non-study entries. `stress_mismatch` fires when either the self-reported mood transitions from positive to negative, or `facial_stress ⩾ 0.5`, giving the facial scan an independent vote alongside the text-based mood check.

Bias classification (`BiasClassifier`). An XGBoost multi-class model, trained on the 11 features, predicts one of five labels (`accurate_perception`, `context_mismatch`, `productivity_overestimation`, `stress_underestimation`, `focus_mismatch`) with a confidence score. Each prediction is explained by reading the booster's per-class `pred_contribs` (a SHAP-style feature-attribution output) and ranking the top three contributing features with their direction (increased/decreased the predicted class) and relative weight — so the student sees *why* an entry was flagged, not just the label. A rule-based layer then adds **additional indicators** the primary prediction may not have covered: e.g. a facial-stress mismatch the text-based mood check missed, high app-switch counts, a location mismatch, or a missing calendar event — each only added when it isn't already the primary bias type, so indicators never duplicate the headline classification.

Perception Alignment Score (`PASCalculator`). A separate, more interpretable 0–100 composite — duration match (30%), focus quality (25%), activity match (20%), location match (15%), calendar match (10%) — mapped to five bands from "Excellent Alignment" (⩾85) to "Severe Bias" (<30). It runs independently of the XGBoost classifier so a student always gets a transparent, formula-based score even if the model's label is wrong or low-confidence.

Orchestration. `BiasDetectionOrchestrator` runs the comparator, then the classifier, then the PAS calculator in sequence and returns the primary bias, additional indicators, PAS score/level/breakdown, and the full comparison payload as one response.

Cross-component mood hand-off. A `pickerMode` flag lets the diary-entry screen (a teammate's `temporal_causal_patterns` component) launch the facial-scan screen and receive the derived mood back via `Navigator.pop(context, mood)`, pre-filling its `MoodSelector` — no shared global state, so each diary entry keeps its own independently-scanned mood. A mapping function translates this component's five mood labels into the diary component's six-label vocabulary, and `Inconclusive` maps to `null` so the student falls back to manual selection rather than a wrong auto-fill.

Evaluation Plan

The PP1 panel's feedback — "research value is limited; need an external evaluation" — most plausibly points at the classifier's training data: `generate_dataset.py` labels its 2,500 synthetic rows with an **author-written, priority-ordered rule chain** (plus 8% injected label noise "tuned to land near a realistic ~90–95% test accuracy"), not real student entries or expert judgement. The reported test accuracy is therefore accuracy against the rules that generated the labels — evidence the model learned the rules, not evidence it detects bias in real students. The plan below is built around closing that specific gap, not just re-stating the existing internal metrics.

1. **Expert-labeled real-data validation set.** Collect 150–200 real diary entries with matching behavioral logs and facial scans from participants outside the project team; have 2–3 independent raters (e.g. supervisors or psychology-adjacent reviewers) label each entry's bias type from the raw comparison data, blind to the model's prediction. Report inter-rater agreement (Cohen's κ) first — if raters cannot agree on the real-world label, the classifier cannot be validated against it either — then the model's accuracy, per-class precision/recall, and confusion matrix against that expert-labeled set, as the external ground truth the synthetic-data accuracy figure is not.
2. **SHAP sanity-check.** For a sample of correctly- and incorrectly-classified real entries, check whether the top SHAP-ranked features match what a human reviewer would point to as the deciding evidence; systematic mismatches would indicate the model learned a spurious correlation from the synthetic generator rather than the intended signal.
3. **Benchmark comparison.** The facial pipeline's confidence/stress thresholds (0.35 reliability cut-off; 0.3/0.5 stress bands) are compared against accuracy ranges reported in [4], [6], and [9], so this implementation's numbers are stated relative to published baselines rather than in isolation.
4. **Threshold sensitivity analysis.** The comparator has several hand-set constants (80 app-switches as the focus-quality ceiling, 35 as the "high switching" rule-layer trigger, 0.5 as the facial-stress cutoff). A small grid search over each, re-run against the expert-

labeled set from (1), would show whether classification quality is robust to these choices or narrowly tuned to the synthetic generator's own ranges.

5. **Integration evaluation.** Because the facial-to-diary mood hand-off is the panel's other explicit ask, its evaluation is separate from classifier accuracy: does the auto-filled mood match what an external participant would have picked manually, and how often is it overridden? A high override rate flags the label-mapping function, independent of FER accuracy.

Limitations. The XGBoost model's feature *distributions* (ranges, correlations) are only as realistic as `generate_dataset.py`'s scenario logic, which was authored to match a now-uncommitted earlier dataset's statistics — a second, independent layer of synthetic-data risk beyond the rule-based labels themselves. Stress is inferred from facial cues alone, not the multimodal ECG/voice fusion used in [8]. GPS location matching and calendar overlap depend on the student maintaining accurate saved reference points and opting into calendar access, so a "mismatch" can reflect stale configuration rather than a real behavioral gap. Finally, even the expert-labeled validation set in (1) is itself a human judgement call, not an objective ground truth — [18]–[20] show self-report bias is pervasive, and the same caution extends to third-party raters interpreting someone else's day from log data alone.

References

IEEE numbering matches the in-text citations above. Entries marked "(authors not individually listed)" had no confirmed author list in the source metadata found — confirm the full author list from the linked page before this goes into the final thesis.

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